

Logical Reconstruction of Arguments in Publications

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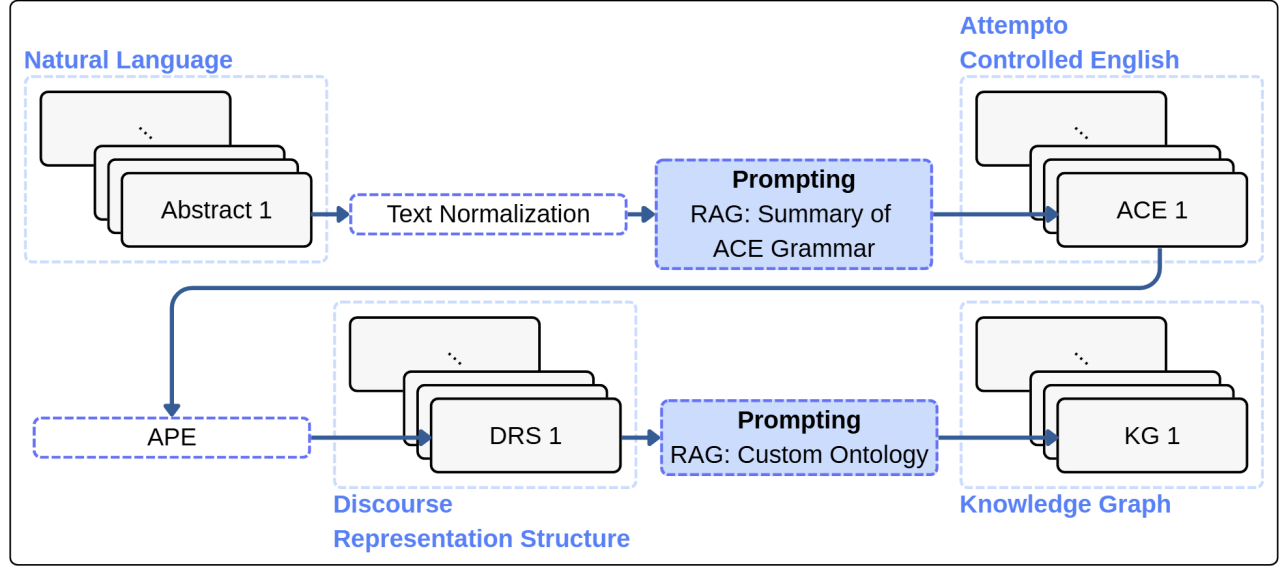


Figure 1: The pipeline for converting biomedical natural language (abstracts from scientific papers) into logical knowledge graphs, consisting of (1) *NL-to-ACE stage*, (2) *ACE-to-DRS stage*, and (3) *DRS-to-KG stage*.

Abstract

Unstructured natural language in scientific publications masks the explicit logical connections between claims, premises, and evidence. Consequently, standard text analysis tools struggle to extract these underlying reasoning structures, hindering comprehensive knowledge synthesis. To address this gap, we present a framework which converts natural language scientific text into logical Knowledge Graphs. Through targeted experiments on a curated corpus, we identify the operational bottlenecks in ontology building and relation extraction. Ultimately, this work offers foundational insights for leveraging Large Language Models to bridge unstructured literature with formal logical representations.

CCS Concepts

• **Computing methodologies** → **Information extraction**; *Ontology engineering*; • **Information systems** → *Graph-based database models*.

Keywords

Scientific Argument Mining, Knowledge Graph Construction, Large Language Models, Logical Knowledge Representation, Information Extraction

1 Introduction

Scientific publications convey complex discourse, including hypotheses, claims, and supporting evidence. However, because these argumentative structures are embedded primarily within unstructured natural language, the inferential links between claims and evidence often remain implicit throughout the text [1]. Consequently, extracting the core logic and systematically evaluating argument validity remains a substantial challenge.

Moreover, terminology varies across different publications. Authors often use different terms to represent the exact same concept depending on the context or scientific discipline. For example, in the medical domain, terms like *cancer*, *malignancy*, and *tumor* are often used interchangeably in different contexts to describe the same underlying condition [13]. This inconsistency creates ambiguity, making it hard to link related arguments across multiple papers.

Additionally, existing text analysis methods, such as entity recognition, have been struggled to capture the underlying reasoning of texts [10]. Extracting individual terms or isolated facts is insufficient to fully comprehend scientific discovery. Instead, analyzing scientific knowledge requires capturing the structural and logical relationships among claims, premises, and arguments.

Given these challenges of logic-aware knowledge representations, we formulate two main research questions:

- How can we design a pipeline for logical reconstruction of arguments in scientific publications?
- What are the key challenges in developing such a pipeline?

In this work, we address these questions by focusing on five primary tasks as follows:

- **Corpus Preparation:** Preparing a small corpus of scientific texts.
- **Pipeline Design:** Designing an overall pipeline for argument reconstruction using a small-scale knowledge graph.
- **Experimental Testing:** Conducting experiments to evaluate each stage of the pipeline.
- **Bottleneck Identification:** Identifying structural limitations and pipeline bottlenecks.
- **Future Directions:** Exploring potential improvements for scaling and automation.

By executing these tasks, we successfully construct the baseline pipeline and identify its operational constraints. Specifically, this work has three main contributions: (1) *pipeline design for logical argument reconstruction*, (2) *challenge identification through preliminary experiments*, and (3) *insights for logical knowledge representation with large language models*.

2 Related Works

2.1 Attempto Controlled English

Attempto Controlled English (ACE) is a controlled natural language which allows automatic translation into formal logic, using a restricted subset of standard English grammar and vocabulary [9]. ACE contains two types of words as follows [9]:

- **Function Words:** Not changeable. (e.g., “If”, “every”, “or” and “is”)
- **Content Words:** Defined freely in six categories. (nouns, proper names, verbs, adjectives, adverbs and preposition)

2.2 Discourse Representation Structure

Discourse Representation Structure (DRS) is a formal semantic representation framework based on Discourse Representation Theory [8], which is a theory for the formal representation of natural discourse [9].

DRSs consist of a domain and of a list of conditions [9]. The domain is a set of discourse referents (i.e. logical variables) and the conditions are a set of first-order logic predicates or nested DRSs [9]. These DRS representations can be translated into various other logic notations [9].

2.3 Attempto Parsing Engine

The Attempto Parsing Engine (APE), which is freely available under an open source license, translates ACE texts into logical format [9]. Its main output is the DRS, serving as a bridge between controlled natural language and formal reasoning systems [9].

2.4 Knowledge graph

A Knowledge Graph (KG) is a semantic network that represents domain knowledge with nodes and edges [16].

A Knowledge Graph can be generally expressed as $G = \{E, R, F\}$, where E is the set of entities, R is the set of relationships defining

connections between entities, and F is the set of facts [16]. Each fact is often defined as a triple formatted as (Entity, Relation, Entity). [16].

2.4.1 Knowledge graph construction using LLM. KG construction using Large Language Models (LLMs) includes works that apply LLMs to address the entity discovery, and relation extraction tasks for KG construction [14]. By leveraging the strong generalizability and natural language processing capabilities of LLMs for knowledge extraction, KGs can overcome their inherent incompleteness and limited language understanding [14].

For such KG construction, LLMs enables entity identification and relation linking into an end-to-end generative process via prompt engineering [19]. Furthermore, providing directly an ontology guides the model to produce domain-compliant triples, mitigating the risk of structural hallucination [12].

3 Methods

As shown in Figure 1, the pipeline converts Natural Language (NL) into a logical Knowledge Graph (KG) via three sequential stages: (1) *NL-to-ACE stage*, (2) *ACE-to-DRS stage*, and (3) *DRS-to-KG stage*.

First, raw scientific text is normalized with PubTator3 [17] entity annotations, and a Large Language Model (LLM) translates it into valid Attempto Controlled English (ACE), guided by Retrieval-Augmented Generation (RAG) based on “ACE 6.7 in a Nutshell” [5] grammar rules. Next, the ACE text is deterministically parsed into Discourse Representation Structures (DRS) via the Attempto Parsing Engine (APE) API [9]. Finally, DRS expressions are mapped into the KG using LLM prompting with RAG to retrieve target structures and constraints from our custom Turtle (TTL) ontology.

3.1 Dataset

To evaluate the proposed pipeline, a total of 10 abstracts related to pancreatic cancer were selected from publications that contain rich pre-existing annotations from PubTator3 [17]. PubTator3 is an advanced web-based text-mining tool provided by the National Center for Biotechnology Information (NCBI), leveraging deep learning architectures to automatically annotate key biomedical entities and relations within PubMed literature [17].

For rigorous testing, ground-truth datasets were manually generated for all four formats of the pipeline: NL, ACE, DRS, and KG. These ground-truth evaluation sets were constructed based on entity and relation annotations from PubTator3 [17], enabling a reliable quality assessment of our pipeline. In particular, the ground-truth KG representations were strictly aligned with the custom ontology introduced in Section 3.2.

Specifically, the dataset contains a total of 158 ACE sentences and an equal number of corresponding DRS representations. For the final stage, the ground-truth KG dataset includes 794 triples. Figure 2 presents the distribution of biomedical entities and relations across the selected abstracts.

3.2 Ontology Generation

To systematically represent medical relations and their underlying logical structures, a domain-specific ontology was constructed using the Resource Description Framework (RDF) and serialized in the TTL format. The ontology comprises a *logical layer* and a

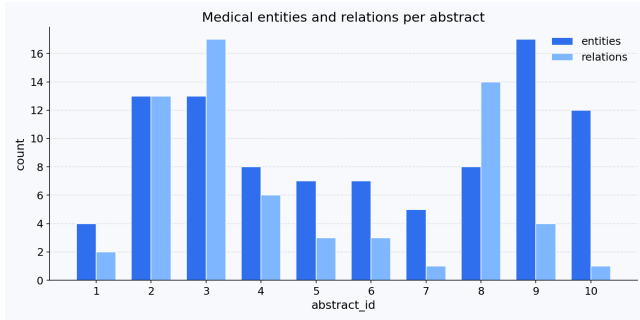


Figure 2: Distribution of medical entities and relations across the abstracts.

medical layer, with their structural mappings detailed in Figure 5 (Appendix).

3.2.1 Logical Layer. The logical layer of the ontology captures the inferential structure of scientific arguments. As shown in Figure 3, it represents formal logical relationships such as implication, negation, disjunction and conjunction of DRSs. This framework establishes a foundation for formal logic operations to reason about and verify scientific arguments in biomedicine.

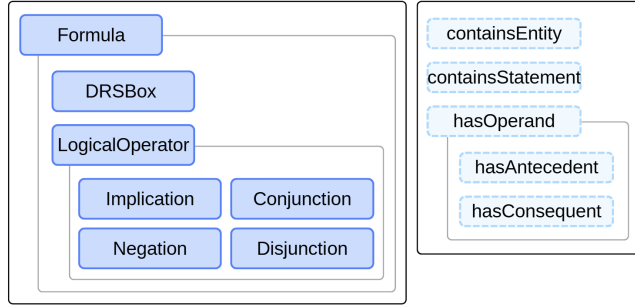


Figure 3: Hierarchical structure of the logical layer in the custom ontology, showing entity (left) and relation hierarchies (right) with subclass groupings.

3.2.2 Medical Layer. To build the logical structures in domain-specific semantics, the medical layer incorporates biomedical entities and relations. As depicted in Figure 4, this layer defines six entity categories (e.g., genes, proteins, and drugs) and twelve relations aligned with PubTator3 annotations [17]. Specifically, relations are grouped into symmetric (e.g., *interact* and *associate*) and asymmetric (e.g., *cause* and *treat*) categories.

3.3 Models and Execution Environment

All prompt-based components of the pipeline were executed within the GWDG (Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen) environment. Inference was conducted using the GWDG SAIA (Scalable AI Accelerator) framework, which provides OpenAI-compatible API access to local HPC clusters [4].

To evaluate prompt engineering strategies, we selected 'Qwen3 30B A3B Instruct 2507' (Qwen) and 'Gemma 4 31B Instruct' (Gemma)

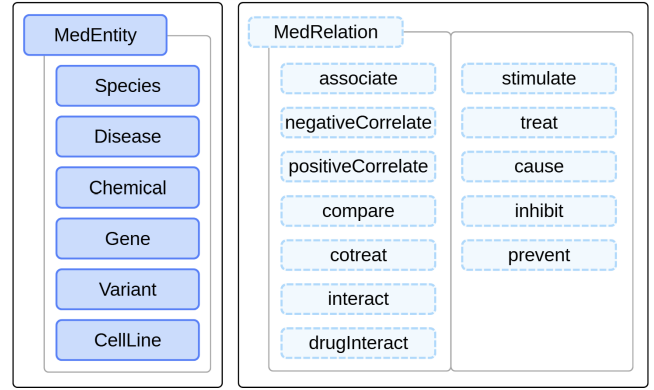


Figure 4: Hierarchical structure of the medical layer in the custom ontology, representing entity (left) with subclass grouping and relation hierarchies (right) categorized by symmetry constraints (left box: symmetric, right box: asymmetric).

as representative models within the ~30B parameter class and comparable native long-context support (~256k tokens) [15, 18]. This pairing enable a comparison between Mixture-of-Experts (MoE) and dense architectures. While Qwen uses 3B active parameters with MoE architecture [18], Gemma activates all 31B parameters as a dense architecture [15]. Evaluating results from both models allows us to examine whether and how prompting results varies across different architectures and methods.

3.4 Prompting with RAG Context

To guide the formal text conversion stages, we utilized zero-shot and one-shot prompting [2] alongside RAG [11]. We integrated RAG via GWDG Arcana [7] to retrieve relevant syntax specifications and domain ontologies to provide structural constraints on the generated outputs.

3.4.1 NL-to-ACE stage. For the first stage, the zero-shot prompt defined the core extraction task, allowed entity prefixes and verbs for the custom ontology, and strict output formatting rules. The one-shot prompt extended this prompt by incorporating an example of a input-output pair. The complete prompt templates are provided in Section A.2 (Appendix).

Notably, to resolve cases where models became trapped in repetitive generation loops, we partitioned the problematic input into smaller segments (typically 2–3 sub-inputs) to complete generation, and manually aggregated the partial outputs.

3.4.2 DRS-to-KG stage. In the last stage, converting DRS to a KG via a single prompt also caused instability with generation loops. We resolved this by decomposing the task into a two-step pipeline: (1) defining simple entity existence (i.e., exists statements) and (2) resolving medical relations yielding multiple triples.

In addition, we optimized stability by processing inputs (DRSs) in batches, with batch sizes set to 5 for step (1) and 2 for step (2). The batch sizes were strictly constrained for zero-shot prompting due to the frequent occurrence of repetitive generation loops under unconstrained settings. Furthermore, RAG was leveraged to supply

our custom ontology for the alignment with predefined entity types and relation predicates. As with the prior stage, zero-shot and one-shot prompts were evaluated, with full details available in Section A.2 (Appendix).

3.5 Evaluation Metrics

Due to the limited scale of our dataset, we employ intrinsic evaluation metrics rather than extrinsic task-based metrics. Intrinsic evaluation assesses output quality [3], enabling us to find structural errors that would otherwise be masked in task performance.

To perform evaluation, predictions are first categorized into True Positives (TP), False Positives (FP), and False Negatives (FN). Specifically, in the *NL-to-ACE stage*, a TP requires both accurate fact extraction and compliance with valid ACE grammar. The predictions that fail either criteria are classified as FP, while omitted ground-truth ACE sentences are FN. In contrast, the *DRS-to-KG stage* operates on a triple-level basis, in which TPs are exact matching TTL-formatted RDF triples, FPs are incorrect or hallucinatory triples, and FNs represent missed target triples [6].

Notably, in the *DRS-to-KG stage*, automated evaluation failed due to ID mismatches with the ground truth caused by batch processing, complexity of comparing symmetric relations, and violations of valid RDF syntax. For these reasons, we manually evaluated the generated outputs. In our manual assessment, a match was credited based on structural and relational validity, ignoring non-essential discrepancies including comments, redundant prefix declarations, and differently assigned IDs.

Based on the counts, we measure the performance via *Precision*, *Recall*, and the F_1 -score [3]. In addition, to evaluate structural limitations, *Hallucination* and *Omission* are measured [6]. Specifically, the metrics are defined as follows:

- **Precision (P) [3]:** Measures the proportion of correctly extracted elements among all predicted outputs.

$$P = \frac{TP}{TP + FP} \quad (1)$$

- **Recall (R) [3]:** Measures the proportion of correctly extracted elements relative to the total ground-truth annotations.

$$R = \frac{TP}{TP + FN} \quad (2)$$

- **F_1 -score [3]:** Represents the harmonic mean of Precision and Recall, providing a balanced overall accuracy score.

$$F_1 = 2 \times \frac{P \times R}{P + R} \quad (3)$$

- **Hallucination [6]:** Quantifies the ratio of model-generated elements that are unsupported by the ground truth.

$$\text{Hallucination} = \frac{FP}{TP + FP} \quad (4)$$

- **Omission [6]:** Quantifies the ratio of ground-truth completely omitted during transformation.

$$\text{Omission} = \frac{FN}{TP + FN} \quad (5)$$

4 Results

4.1 NL-to-ACE Stage

As illustrated in Table 1, the overall results demonstrate that one-shot prompting outperforms zero-shot prompting across all metrics, using ‘Gemma 4 31B Instruct’ (Gemma) and one-shot configuration achieving the highest performance overall. However, while prompting strategies effectively reduce errors across both models, these gains remain moderate, staying below 0.1 without introducing drastic performance improvements.

Notably, the two models exhibit distinct error characteristics. Based on a manual inspection of errors, Gemma’s primary weakness under zero-shot prompting was the omission of basic entity definitions, which was mitigated by incorporating a single one-shot example. In contrast, ‘Qwen3 30B A3B Instruct 2507’ (Qwen) had severe hallucination issues, resulting in significantly lower precision. Regarding omission errors, both models perform within a comparable range (ranging from 0.3 to 0.5), though Gemma still outperforms Qwen.

Based on the specified results in Tables 3– 6 (Appendix), we further examined extraction performance regarding the scale and composition of the data. Given the total number of entities and relations, the abstracts were divided into three groups: small (< 10 elements; IDs 1 and 7), medium (10 – 19 elements; IDs 4, 5, 6, and 10), and large (≥ 20 elements; IDs 2, 3, 8, and 9). For Gemma, mean F_1 score decreased consistently as the data scale increased, from 0.817 for the small group to 0.716 for the medium group and 0.492 for the large group in the zero-shot setting. The same overall trend was observed with one-shot prompting (0.962, 0.716, and 0.592, respectively). Qwen also showed a decrease of mean F_1 score in zero-shot setting. However, its one-shot results were not consistent, with the large group (0.260) slightly outperforming the medium group (0.217).

Regarding the composition of entities and relations, abstracts with relatively more entities (IDs 9 and 10) showed consistently high recall across both models. For Gemma, the entity-heavy cases achieved a mean recall of 0.890 in zero-shot and 0.938 in one-shot. Qwen showed the same pattern, although with lower precision and higher hallucination rates. In contrast, the relation-heavy cases (IDs 3 and 8) exhibited relatively lower recall. For Gemma, a mean recall was 0.397 in zero-shot and 0.414 in one-shot, while Qwen showed a stronger effect, with a mean recall of 0.124 in zero-shot and 0.308 in one-shot.

4.2 DRS-to-KG Stage

As presented in Table 2, converting Discourse Representation Structures (DRS) to a Knowledge Graph (KG) exhibits a dramatic performance improvement when transitioning from zero-shot to one-shot prompting across both models. Specifically, Gemma’s F_1 -score jumps from 0.221 to 0.950. Moreover, Qwen outperforms Gemma in both settings, even achieving perfect scores (1.000) across all metrics in the one-shot setup. These results demonstrate that while in-context guidance is highly effective for both models, Qwen possesses a superior architectural capacity for structured DRS-to-KG conversion.

To investigate the sources of extraction errors, we mainly focused on zero-shot results as the performance under one-shot prompting

Table 1: NL-to-ACE Results

Model	Prompting	TP	FP	FN	Precision	Recall	F1 Score	Hallucination	Omission
Gemma	zero-shot	87	62	71	0.584	0.551	0.567	0.416	0.449
Gemma	one-shot	97	53	61	0.647	0.614	0.630	0.353	0.386
Qwen	zero-shot	72	354	86	0.169	0.456	0.247	0.831	0.544
Qwen	one-shot	83	380	75	0.179	0.525	0.267	0.821	0.475

Table 2: DRS-to-KG stage Results

Model	Prompting	TP	FP	FN	Precision	Recall	F1 Score	Hallucination	Omission
Gemma	zero-shot	189	725	605	0.207	0.238	0.221	0.793	0.762
Gemma	one-shot	755	40	39	0.950	0.950	0.950	0.050	0.049
Qwen	zero-shot	253	541	541	0.319	0.319	0.319	0.681	0.681
Qwen	one-shot	794	0	0	1.000	1.000	1.000	0.000	0.000

provided too few error instances for a qualitative analysis. Based on the Tables 7 and 9 (Appendix), we again grouped the abstracts by categorizing them into small (< 10 elements; IDs 1 and 7), medium ($10 - 19$ elements; IDs 4, 5, 6, and 10), and large (≥ 20 elements; IDs 2, 3, 8, and 9) groups. Unlike the first stage evaluation, the results do not show a consistent relationship between total data scale and performance. For Gemma, the small group actually produced the lowest F_1 score (0.054), compared with 0.249 for the medium group and 0.237 for the large group. Qwen showed a gradual decrease from 0.398 to 0.347 and 0.340, but the difference was relatively small.

The composition of entities and relations provides a more informative distinction. Abstracts with high entity density (IDs 9 and 10) achieved higher zero-shot performance than the overall average for both models, with lower hallucination and omission. Gemma achieved an average F_1 score of 0.381 in these cases, compared with 0.221 overall, while Qwen achieved 0.686 compared with 0.319 overall. In contrast, the relation-heavy cases (IDs 3 and 8) exhibited relatively lower performance. Gemma achieved an average F_1 of 0.182, and Qwen showed a similar pattern, with an average F_1 score of 0.267.

5 Discussion

5.1 Interpretation of Findings

Overall, the empirical findings demonstrate that one-shot prompting is essential for converting unstructured natural language into logical representations.

5.1.1 NL-to-ACE stage. In the *NL-to-ACE stage*, ‘Gemma 4 31B Instruct’ (Gemma) outperformed ‘Qwen3 30B A3B Instruct 2507’ (Qwen), as Qwen exhibited a noticeably higher frequency of hallucinations. This tendency contributed to Qwen’s frequent generation loops. Based on the irrelevant text accumulated within its context window, the model continuously attended to its own hallucinated outputs, ultimately trapping itself in repetitive loops.

Also, while relation-heavy abstracts generally lead to higher omission rates, total entity or relation counts alone do not fully explain performance. For example, ID 3 contains 17 relations, the highest relation count among the abstracts, but Gemma achieves a

higher F_1 score of 0.680 than ID 2 (0.244) and ID 8 (0.278) in the zero-shot setting. Thus, performance depends on the entity-to-relation composition rather than total relation count alone.

Overall, the results indicate that both abstract size and entity-relation composition are relevant factors in extraction difficulty. Gemma appears more robust to these factors than Qwen, while Qwen consistently exhibits high hallucination rates. Also, one-shot prompting improves Gemma more consistently, whereas its effect on Qwen is comparatively limited, although it shows a notable improvement for the relation-heavy cases.

5.1.2 DRS-to-KG stage. In the *DRS-to-KG stage*, the poor performance under zero-shot prompting was primarily caused by severe syntax violations rather than semantic misunderstanding. The models consistently failed to follow Turtle (TTL) grammar rules, frequently missing required prefix declarations. Remarkably, one-shot prompting completely resolves these syntax errors. This dramatic improvement stems from the deterministic nature of the task, which requires structural mapping, once generation loops are controlled by using smaller batch sizes. Consequently, providing a single example proved sufficient to guide the model toward generating valid triples. Additionally, the hallucination and omission under one-shot prompting in Gemma show a nearly 1 : 1 correspondence, where errors are from medical relation predictions rather than formal structural failures.

Regarding data scale, focusing primarily on the zero-shot setting, the performance differences across abstract sizes were minimal. This small variance is likely related to processing inputs in batches regardless of their length. Therefore, the current results do not provide sufficient evidence that larger abstracts are inherently more difficult for zero-shot extraction.

Furthermore, across both models, relation-heavy abstracts are related to poorer performance, whereas entity-heavy abstracts tend to have fewer errors. However, the absolute number of entities or relations alone does not consistently explain the observed performance. Specifically, performances of the absolute entity-heavy abstracts (IDs 2, 3, 9, and 10) or the absolute relation-heavy abstracts (IDs 2, 3, and 8) does not show a consistent pattern relative to the overall results. Similar to the *NL-to-ACE stage*, these observations

imply that the relative composition of entities and relations may be more informative than their absolute counts.

Nevertheless, the dataset size is small in this work, so these findings should be interpreted as observed tendencies rather than statistically established effects.

5.2 Limitations

While our pipeline shows promising performance in converting biomedical scientific literature into structured Knowledge Graphs (KGs), several limitations remain.

First, reliance on an external API for prompting introduces severe operational instability. Frequent timeouts disrupt continuous processing, and models occasionally enter repetitive generation loops, which are caused by listing multiple similar entities in the *NL-to-ACE stage* or generating up to eight triples per relation in the *DRS-to-KG stage*, requiring manual intervention. Consequently, the pipeline cannot process all inputs at once or update relations dynamically in real time. Requiring static input sets with manual inspection limits both scalable deployment and real-time argument extraction.

Next, the pipeline heavily relies on PubTator3 [17] for entity and relation annotations. Specifically, manual error inspection shows that a portion of the incorrect predictions originated from noisy or inconsistent PubTator3 annotations [17] rather than model failure alone. Furthermore, since it is an AI-driven tool rather than a logical reasoning system, its outputs lack logical grounding. Therefore, annotation errors and structural inconsistencies are frequently observed, which directly degrade the quality of the reconstructed Knowledge Graph.

Lastly, the pipeline currently uses a restricted ontology for entities and relations. This rigid framework cannot fully capture complex scientific arguments or domain-specific terminology that are not included in the predefined schema. Also, existing relation types from PubTator3 [17] are poorly differentiated, with most links broadly classified as *"associate"*. As an overly generalized term, *"associate"* prevents distinguishing specific medical relation types, fundamentally limiting formal logic inference. As a result, critical contextual semantics are oversimplified during the knowledge graph mapping process. More importantly, even when the current ontology represents individual facts in formal logic, actual logical operations remain restricted because the composition rules regarding how these relations interact to yield inferred conclusions have not yet been defined.

5.3 Future Works

Building upon the insights and limitations identified in this work, we propose several key directions for future research.

First, to mitigate API instability, future work should focus on prompt optimization and model fine-tuning to stabilize Attempto Controlled English (ACE) and KG outputs. Achieving this output consistency is a critical prerequisite for automating the evaluation process. Once evaluation is automated, the entire pipeline can be seamlessly scaled while maintaining high extraction quality, ultimately enabling real-time, dynamic KG construction without static constraints.

Second, high-quality, human-verified annotations are necessary for the future research to address PubTator3's annotation errors. Ground-truth supervision from domain experts will serve as more reliable data to refine entity recognition and relation extraction than the AI-driven tool.

Third, the predefined schema should be expanded to explain a broader range of biomedical concepts with logical operations. Defining systematic and rich relation types will allow the graph to go beyond simple factual associations and capture semantic-level logical reasoning. Specifically, by defining how multiple relations interact to yield valid semantic conclusions, the graph will move beyond static factual assertions to support complex, multi-hop logical reasoning.

Lastly, the pipeline can be scaled up to construct a comprehensive, domain-specific KG tailored for multi-document logical reasoning. By linking structured arguments across vast collections of biomedical literature, this broader framework will enable automated cross-document validation, and the discovery of novel implicit relations. Furthermore, such a logical KG will facilitate automated contradiction detection across conflicting medical studies, laying the foundation for automated hypothesis generation and synthesis in complex biomedical domains.

6 Conclusion

In this work, we presented a pipeline for reconstructing logical arguments and biomedical claims from scientific literature into formal Knowledge Graphs (KGs). By constructing KGs leveraging Large Language Models (LLMs), we demonstrated that LLM-driven logical extraction is reliable when properly constrained. Our empirical evaluations revealed that while the *NL-to-ACE stage* is sensitive to sentence complexity and annotation noise, the *DRS-to-KG stage* achieves drastically high precision with one-shot prompting.

The main contributions of this work are summarized as follows. First, we develop an initial framework designed to logically reconstruct complex arguments from scientific publications. The proposed architecture systematically integrates entity and relation extraction in Attempto Controlled English (ACE) format, ACE-to-DRS conversion via APE, and KG generation from Discourse Representation Structures (DRSs) inputs to transform unstructured textual claims into machine-readable structures.

Second, by evaluating the pipeline on a small-scale corpus, this work identifies key operational bottlenecks and limitations. The experiments highlight critical challenges in the annotation quality, the constraints of fixed representation based on the custom ontology, and the current boundaries of automated logical reasoning capabilities.

Ultimately, this work provides essential insights toward logic-aware knowledge systems. By demonstrating that Large Language Models can reliably connect unstructured scientific text with formal logical representations, our work establishes a practical framework for expanding automated literature analysis into precise, logic-based knowledge representation.

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A Detailed Methodology

A.1 Custom Ontology

To support conversion from DRS to Knowledge Graphs, our custom ontology aligns formal logic structures with domain-specific medical concepts. Figure 5 details the mapping rules and structures established between the *logical layer* and the *medical layer*.

A.2 Prompts

A.2.1 NL to ACE stage. We employ prompt-based generation to translate normalized natural language (NL) texts into Attempto Controlled English (ACE). Below is the system prompt used for the zero-shot setting.

You are an expert in Pancreatic Cancer and Attempto Controlled English (ACE).
Convert the given biomedical natural language text into valid ACE sentences.

Goal
Extract only concrete biomedical facts from the input.
For each fact, write either an existence sentence or a binary medical relation sentence in ACE.

Allowed Entity Prefixes
Use only entity identifiers that already appear in the input with one of these prefixes: Species_, Disease_, Chemical_, Gene_, Variant_, CellLine_.
In ACE output, every entity MUST use the p: prefix, e.g., p:Disease_MESH_D021441.

Allowed Verbs

- **Symmetric:** v:associate_with, negatively v:correlate_with, positively v:correlate_with, v:be_compared_with, v:cotreat_with, v:interact_with, v:interact_as_drug_with
- **Asymmetric:** v:cause, v:inhibit, v:treat, v:stimulate, v:prevent

Rules

- (1) Return ONLY raw ACE sentences.
- (2) Keep ONLY two categories of facts: p:Entity exists . or p:Entity1 v:relation p:Entity2 .
- (3) DELETE meta-discourse, summaries, overviews, and non-factual text.
- (4) IGNORE and DROP any entity or relation that is not in the allowed lists above.
- (5) Avoid overly long or complicated sentences. Split complex facts into short ACE sentences.
- (6) Use and, or, if ... then ..., it is not the case that for logical operations when needed.
- (7) Avoid ambiguous pronouns such as their or its. Repeat the entity identifier explicitly.
- (8) After all extractable facts are covered, stop immediately.

Now, convert the following input text into valid ACE sentences:

To improve compliance with ACE syntax rules, the one-shot prompt incorporates an example of input text and target ACE statements:

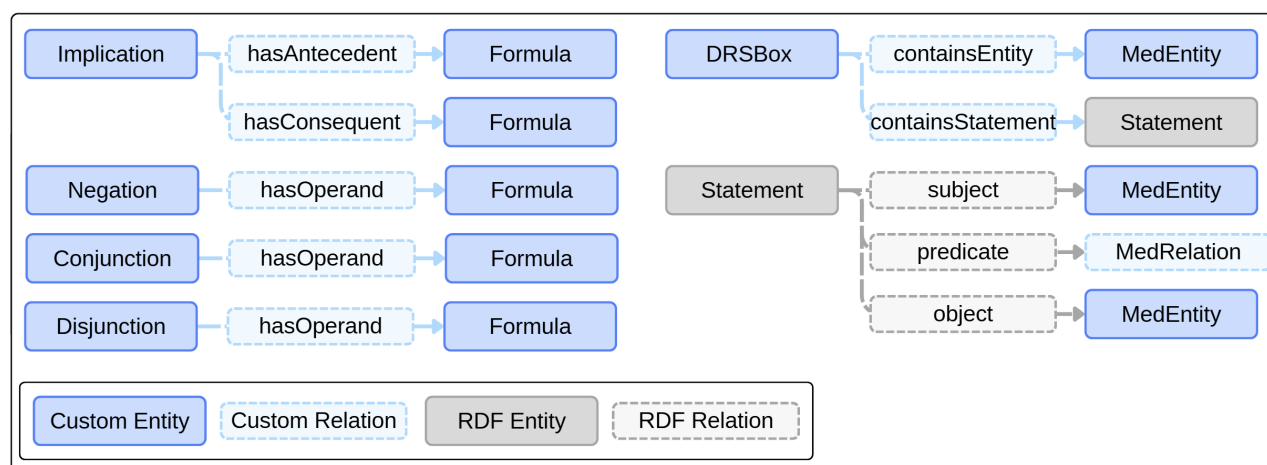


Figure 5: Structural Mapping Rules between the Logical and Medical Layers of the Custom Ontology.

[... Continued from Zero-shot Prompt ...]

Minimal Example

Input: "The influence of cigarette smoking..."

Output:

p:Chemical_example_01 exists .

p:Disease_example_01 exists .

p:Chemical_example_01 v:cause p:Disease_example_01 .

[...]

Now, convert the following input text into valid ACE sentences:

A.2.2 DRS to KG stage. The 'Discourse Representation Structures (DRS) to Knowledge Graph (KG)' mapping is executed via a two-step pipeline: (1) defining simple entity existence, and (2) resolving medical relations into target Knowledge Graph triples. The system prompts used for each step across zero-shot and one-shot settings are presented below.

Below is the system prompt for Step 1 used in the zero-shot setting.

You are an expert in Pancreatic Cancer, Discourse Representation Structures (DRS), and Knowledge Graph engineering. Convert the given DRS input into valid Turtle/RDF.

Goal

The input for this prompt contains only exist predicates.

For each input line, create the entity type triple and exactly one DRS box.

Allowed Classes

Use only these medical entity classes for `rdf:type`: `lkg:Species`, `lkg:Disease`, `lkg:Chemical`, `lkg:Gene`, `lkg:Variant`, `lkg:CellLine`. Use `lkg:DRSBox` for the box.

Allowed Properties

Use only: `rdf:type`, `lkg:containsEntity`.

Rules

- (1) Return ONLY raw Turtle/RDF.
- (2) Always use `rdf:type`, never `a`.
- (3) For each exist input line, create exactly: one entity type triple, one `lkg:DRSBox`, and one `lkg:containsEntity`.
- (4) Do not create any `rdf:Statement`.
- (5) Do not create any `lkg:containsStatement`.
- (6) Do not merge multiple input lines into one `DRSBox`.
- (7) Every unique RDF triple must appear exactly once.
- (8) After all input lines are covered exactly once, stop immediately.

Now, convert the following input text into valid Knowledge Graph:

To improve performance, the one-shot prompt for Step 1 incorporates an example pair of an input DRS expression and target *exists* statements:

```
[... Continued from zero-shot Prompt ...]

### Minimal Example
Input: "[A] predicate(A,exist,named(Cheical_example_01))-1/4"
Output:
lkg:Chemical_example_01 rdf:type lkg:Chemical .
lkg:DRS_01 rdf:type lkg:DRSBox ;
lkg:containsEntity lkg:Chemical_example_01 .

Now, convert the following input text into valid Knowledge Graph:
```

Below is the system prompt for Step 2 used in the zero-shot setting to resolve medical relations into Knowledge Graph triples.

```
You are an expert in Pancreatic Cancer, Discourse Representation Structures (DRS), and Knowledge Graph engineering. Convert the given DRS input into valid Turtle/RDF.

### Goal
The input for this prompt contains only binary relation predicates such as treat or associate_with.
For each input line, create exactly one DRS box and exactly one RDF statement.

### Allowed Classes
Use only: lkg:DRSBox, rdf:Statement.
Entity names in the input already identify the medical entities. Use the ontology-defined type or relation names only when needed in RDF triples.

### Allowed Properties
Use only: rdf:type, lkg:containsEntity, lkg:containsStatement, rdf:subject, rdf:predicate, rdf:object.

### Rules
(1) Return ONLY raw Turtle/RDF.
(2) Always use rdf:type, never a.
(3) For each relation input line, create exactly: one lkg:DRSBox, two lkg:containsEntity, one lkg:containsStatement, one rdf:Statement, one rdf:subject, one rdf:predicate, and one rdf:object.
(4) Do not create entity type triples in this prompt.
(5) Do not merge multiple input facts into one DRSBox.
(6) Do not create multiple DRSBoxes or multiple rdf:Statements for the same input line.
(7) Every unique RDF triple must appear exactly once.
(8) After all input lines are covered exactly once, stop immediately.

Now, convert the following input text into valid Knowledge Graph:
```

To improve performance, the one-shot prompt for Step 2 incorporates an example pair of an input DRS expression and target relation triples:

```
[... Continued from zero-shot Prompt ...]

### Minimal Example
Input: "[A] predicate(A,treat,named(Cheical_example_01),named(Disease_example_01))-1/6"

Output:
lkg:DRS_01 rdf:type lkg:DRSBox ;
lkg:containsEntity lkg:Chemical_example_01 ,
lkg:Disease_example_01 ;
[...]
rdf:object lkg:Disease_example_01 .

Now, convert the following input text into valid Knowledge Graph:
```

B Detailed Results across Datasets

B.1 NL to ACE Stage

This section presents the dataset-level evaluation results for the *NL-to-ACE stage*. We evaluate both 'Gemma 4 31B Instruct' (Gemma) and 'Qwen3 30B A3B Instruct 2507' (Qwen) models under zero-shot and one-shot settings. The detailed results are summarized in Tables 3 to 6.

B.2 DRS to KG

This section presents the dataset-level evaluation results for the *DRS-to-KG stage*. We evaluate both Gemma and Qwen models under zero-shot and one-shot settings. The detailed results are summarized in Tables 7 to 10.

C Online Resources

The code, dataset, and system prompts used in this work are publicly available on GitHub at <https://github.com/JihooChung/logicalKG>.

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Table 3: NL-to-ACE stage: Gemma Zero-shot Results

abstract id (# of entities + # of relations)	TP	FP	FN	Precision	Recall	F1 Score	Hallucination	Omission
1 (4+2)	4	0	2	1.0	0.667	0.8	0.0	0.333
2 (13+13)	5	10	21	0.333	0.192	0.244	0.667	0.808
3 (13+17)	17	3	13	0.85	0.567	0.68	0.15	0.433
4 (8+6)	10	4	4	0.714	0.714	0.714	0.286	0.286
5 (7+3)	7	6	3	0.538	0.7	0.609	0.462	0.3
6 (7+3)	10	2	0	0.833	1.0	0.909	0.167	0.0
7 (5+1)	5	1	1	0.833	0.833	0.833	0.167	0.167
8 (8+14)	5	9	17	0.357	0.227	0.278	0.643	0.773
9 (17+4)	18	8	3	0.692	0.857	0.766	0.308	0.143
10 (12+1)	12	13	1	0.48	0.923	0.632	0.52	0.077
total	93	56	65	0.624	0.589	0.606	0.376	0.411

Table 4: NL-to-ACE stage: Gemma One-shot Results

abstract id (# of entities + # of relations)	TP	FP	FN	Precision	Recall	F1 Score	Hallucination	Omission
1 (4+2)	6	0	0	1.0	1.0	1.0	0.0	0.0
2 (13+13)	10	6	16	0.625	0.385	0.476	0.375	0.615
3 (13+17)	18	1	12	0.947	0.6	0.735	0.053	0.4
4 (8+6)	10	4	4	0.714	0.714	0.714	0.286	0.286
5 (7+3)	7	6	3	0.538	0.7	0.609	0.462	0.3
6 (7+3)	10	2	0	0.833	1.0	0.909	0.167	0.0
7 (5+1)	6	1	0	0.857	1.0	0.923	0.143	0.0
8 (8+14)	5	8	17	0.385	0.227	0.286	0.615	0.773
9 (17+4)	20	5	1	0.8	0.952	0.87	0.2	0.048
10 (12+1)	12	13	1	0.48	0.923	0.632	0.52	0.077
total	104	46	54	0.693	0.658	0.675	0.307	0.342

Table 5: NL-to-ACE stage: Qwen Zero-shot Results

abstract id (# of entities + # of relations)	TP	FP	FN	Precision	Recall	F1 Score	Hallucination	Omission
1 (4+2)	6	15	0	0.286	1.0	0.444	0.714	0.0
2 (13+13)	12	57	14	0.174	0.462	0.253	0.826	0.538
3 (13+17)	2	32	28	0.059	0.067	0.062	0.941	0.933
4 (8+6)	8	23	6	0.258	0.571	0.356	0.742	0.429
5 (7+3)	2	15	8	0.118	0.2	0.148	0.882	0.8
6 (7+3)	7	56	3	0.111	0.7	0.192	0.889	0.3
7 (5+1)	5	22	1	0.185	0.833	0.303	0.815	0.167
8 (8+14)	4	32	18	0.111	0.182	0.138	0.889	0.818
9 (17+4)	17	62	4	0.215	0.81	0.34	0.785	0.19
10 (12+1)	9	40	4	0.184	0.692	0.29	0.816	0.308
total	72	354	86	0.169	0.456	0.247	0.831	0.544

Table 6: NL-to-ACE stage: Qwen One-shot Results

abstract id (# of entities + # of relations)	TP	FP	FN	Precision	Recall	F1 Score	Hallucination	Omission
1 (4+2)	6	10	0	0.375	1.0	0.545	0.625	0.0
2 (13+13)	10	36	16	0.217	0.385	0.278	0.783	0.615
3 (13+17)	13	40	17	0.245	0.433	0.313	0.755	0.567
4 (8+6)	8	28	6	0.222	0.571	0.32	0.778	0.429
5 (7+3)	2	33	8	0.057	0.2	0.089	0.943	0.8
6 (7+3)	7	57	3	0.109	0.7	0.189	0.891	0.3
7 (5+1)	5	16	1	0.238	0.833	0.37	0.762	0.167
8 (8+14)	4	30	18	0.118	0.182	0.143	0.882	0.818
9 (17+4)	18	79	3	0.186	0.857	0.305	0.814	0.143
10 (12+1)	10	51	3	0.164	0.769	0.27	0.836	0.231
total	83	380	75	0.179	0.525	0.267	0.821	0.475

Table 7: DRS-to-KG stage: Gemma Zero-shot Results

abstract id (# of entities + # of relations)	TP	FP	FN	Precision	Recall	F1 Score	Hallucination	Omission
1 (4+2)	2	30	26	0.063	0.071	0.067	0.938	0.929
2 (13+13)	43	120	100	0.264	0.301	0.281	0.736	0.699
3 (13+17)	41	158	134	0.206	0.234	0.219	0.794	0.766
4 (8+6)	15	67	57	0.183	0.208	0.195	0.817	0.792
5 (7+3)	9	44	36	0.170	0.200	0.184	0.830	0.800
6 (7+3)	8	47	37	0.145	0.178	0.160	0.855	0.822
7 (5+1)	1	26	22	0.037	0.043	0.040	0.963	0.957
8 (8+14)	21	135	115	0.135	0.154	0.144	0.865	0.846
9 (17+4)	27	68	56	0.284	0.325	0.303	0.716	0.675
10 (12+1)	22	30	22	0.423	0.500	0.458	0.577	0.500
total	189	725	605	0.207	0.238	0.221	0.793	0.762

Table 8: DRS-to-KG stage: Gemma One-shot Results

abstract id (# of entities + # of relations)	TP	FP	FN	Precision	Recall	F1 Score	Hallucination	Omission
1 (4+2)	28	0	0	1.000	1.000	1.000	0.000	0.000
2 (13+13)	130	13	13	0.909	0.909	0.909	0.091	0.091
3 (13+17)	171	4	4	0.977	0.977	0.977	0.023	0.023
4 (8+6)	69	3	3	0.958	0.958	0.958	0.042	0.042
5 (7+3)	42	3	3	0.933	0.933	0.933	0.067	0.067
6 (7+3)	42	3	3	0.933	0.933	0.933	0.067	0.067
7 (5+1)	23	0	0	1.000	1.000	1.000	0.000	0.000
8 (8+14)	124	12	12	0.912	0.912	0.912	0.088	0.088
9 (17+4)	83	2	0	0.976	1.000	0.988	0.024	0.000
10 (12+1)	43	0	1	1.000	0.977	0.989	0.000	0.023
total	755	40	39	0.950	0.951	0.950	0.050	0.049

Table 9: DRS-to-KG stage: Qwen Zero-shot Results

abstract id (# of entities + # of relations)	TP	FP	FN	Precision	Recall	F1 Score	Hallucination	Omission
1 (4+2)	4	24	24	0.143	0.143	0.143	0.857	0.857
2 (13+13)	39	104	104	0.273	0.273	0.273	0.727	0.727
3 (13+17)	42	133	133	0.240	0.240	0.240	0.760	0.760
4 (8+6)	14	58	58	0.194	0.194	0.194	0.806	0.806
5 (7+3)	6	39	39	0.133	0.133	0.133	0.867	0.867
6 (7+3)	11	34	34	0.244	0.244	0.244	0.756	0.756
7 (5+1)	15	8	8	0.652	0.652	0.652	0.348	0.348
8 (8+14)	40	96	96	0.294	0.294	0.294	0.706	0.706
9 (17+4)	46	37	37	0.554	0.554	0.554	0.446	0.446
10 (12+1)	36	8	8	0.818	0.818	0.818	0.182	0.182
total	253	541	541	0.319	0.319	0.319	0.681	0.681

Table 10: DRS-to-KG stage: Qwen Zero-shot Results

abstract id (# of entities + # of relations)	TP	FP	FN	Precision	Recall	F1 Score	Hallucination	Omission
1 (4+2)	28	0	0	1.000	1.000	1.000	0.000	0.000
2 (13+13)	143	0	0	1.000	1.000	1.000	0.000	0.000
3 (13+17)	175	0	0	1.000	1.000	1.000	0.000	0.000
4 (8+6)	72	0	0	1.000	1.000	1.000	0.000	0.000
5 (7+3)	45	0	0	1.000	1.000	1.000	0.000	0.000
6 (7+3)	45	0	0	1.000	1.000	1.000	0.000	0.000
7 (5+1)	23	0	0	1.000	1.000	1.000	0.000	0.000
8 (8+14)	136	0	0	1.000	1.000	1.000	0.000	0.000
9 (17+4)	83	0	0	1.000	1.000	1.000	0.000	0.000
10 (12+1)	44	0	0	1.000	1.000	1.000	0.000	0.000
total	794	0	0	1.000	1.000	1.000	0.000	0.000